

Differentiation

Product Rule

Differentiation – Product Rule

- The **Product Rule** is used when differentiating two or more functions that are being multiplied together
- It is used to determine the derivative of the product of two functions

Product Rule

If $y = uv$ where u and v are functions of x , then

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Differentiation – Product Rule

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

EXAMPLE 1

- Differentiate:

$$y = (4x^3 - x)(5 - 5x)$$
$$u = 4x^3 - x$$
$$\frac{du}{dx} = 12x^2 - 1$$
$$v = 5 - 5x$$
$$\frac{dv}{dx} = -5$$

Differentiation – Product Rule

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

EXAMPLE 1

$$\begin{aligned} \frac{dy}{dx} &= \overbrace{(4x^3 - x)}^u \overbrace{(-5)}^{\frac{dv}{dx}} + \overbrace{(5 - 5x)}^v \overbrace{(12x^2 - 1)}^{\frac{du}{dx}} \\ \frac{dy}{dx} &= -20x^3 + 5x + 60x^2 - 5 - 60x^3 + 5x \\ \frac{dy}{dx} &= -80x^3 + 10x + 60x^2 - 5 \end{aligned}$$

Differentiation – Product Rule

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

EXAMPLE 1

$$\frac{dy}{dx} = -80x^3 + 10x + 60x^2 - 5$$

- The result is often written in descending order

$$\frac{dy}{dx} = -80x^3 + 60x^2 + 10x - 5$$

- This could be factorised to

$$\frac{dy}{dx} = 5(-16x^3 + 12x^2 + 2x - 1)$$

Differentiation – Product Rule

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

EXAMPLE 2

- Find the value of $\frac{dy}{dx}$ if $x = 1$

$$\begin{aligned} \frac{dy}{dx} &= -80x^3 + 60x^2 + 10x - 5 \\ \frac{dy}{dx} &= -80(1^3) + 60(1^2) + 10(1) - 5 \\ \frac{dy}{dx} &= -80 + 60 + 10 - 5 \\ \frac{dy}{dx} &= -15 \end{aligned}$$